NIKE
AIR FORCE 1 LOW



REVERSE TECH PACK

BY: AIDAN COPELAND

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AIR FORCE 1 LOW

Fall '83



Model Name	NIKE AIR FORCE 1 LOW
Forecast	25,000 PAIRS - U.S. ESTIMATE
Gender	MEN'S
Size Range	MEN'S U.S. 7-13 (W/ HALF SIZES) 14-15 - SUBJECT TO DEMAND
Sample Size	MEN'S U.S. - SIZE 9
Retail/Wholesale Price	$65.00/$32.50
Target FOB	$13.00
Life Cycle	12-18 MONTHS
Distribution Plan	U.S. ATHLETIC FOOTWEAR REATILERS AND SPORTING GOODS STORES

Objective

Extend Nike Air basketball cushioning into a low-cut option for players who prefer a lower collar. The low-top may also promote increased casual wear of the popular Air Force 1 silhouette.

Target Consumer

Target competitive high school, collegiate, and recreational basketball players seeking premium cushioning, court traction, and reliable support. Appeal to athletes who prioritize performance and mobility, with secondary interest from consumers seeking versatile athletic footwear for casual wear.



Last



New - low-top variation of the existing Air Force 1 high-top last

Challenges

- Cost
- Weight
- Ankle Support
- Fit and Heel Retention

Heroes



- NIKE Air Force 1 Silhouette
- NIKE Air Cushioning System
- Low-Top Design
- Leather Support Upper
- Circular Unit-sole Traction Pattern

Key Product Features

1. Premium leather upper with reinforced panel construction
2. Low-cut padded collar to accommodate ankle mobility
3. Perforated leather toe panel to improve ventilation
4. Traditional adjustable lace closure
5. Encapsulated Nike Air-Sole cushioning

Material Information

Upper Materials:
- Full-Grain Leather (Vamp, Quarter Panels & Overlays)
- Perforated Leather (Toe Box)
- Leather (Nike Swoosh Overlays)
- Reinforcement Materials (Toe & Heel)

Midsole Materials:
- Rubber Cupsole Construction
- Encapsulated Nike Air-Sole Cushioning Unit
- Foam Cushioning / Filler (Exact Composition Unconfirmed)

Outsole Materials:
- Durable Molded Rubber
- Concentric Circular Pivot Traction Pattern
- Textured Perimeter Traction

Tongue Materials:
- Leather / Textile Outer Construction
- Foam Padding Interior
- Woven Nike Logo Label

Other Materials:
- Textile Collar Lining with Foam Padding
- Flat Woven Textile Laces
- Cushioned Sockliner with Textile Top Cover
- Leather Heel Tab with Nike Branding
- Synthetic Thread – Upper Panel Stitching

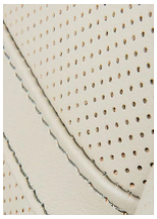
Inspiration

1.

2.



3.



4.



5.



Colorways

Launch Colorway
- White/Neutral Gray

Secondary Launch Colorway
- Team Color Options (pending verification)

Competitors

$70

adidas



Adidas Top Ten

$35

CONVERSE



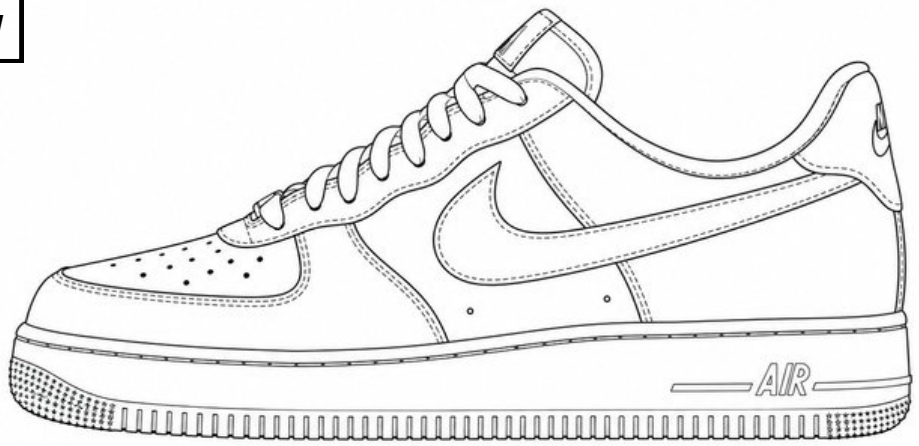
Converse Pro Leather

NIKE Air Force 1 Low

Blank View

MODEL:	AIR FORCE 1 LOW	SHEET:	1 OF 8
SEASON:	FALL 1983	REV:	1
SIZE:	U.S. M9	DATE:	2/6/82

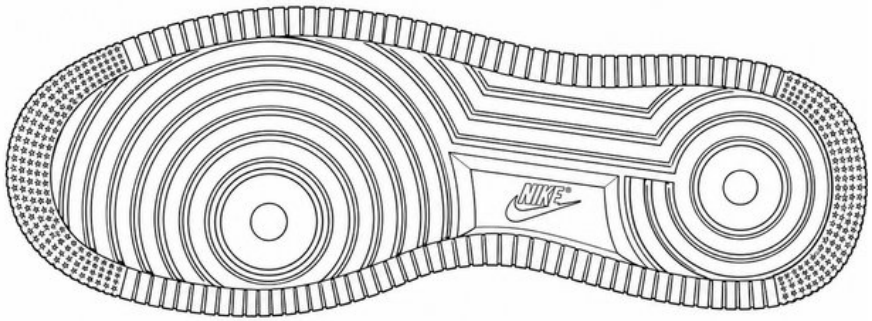
LATERAL VIEW



HEEL VIEW



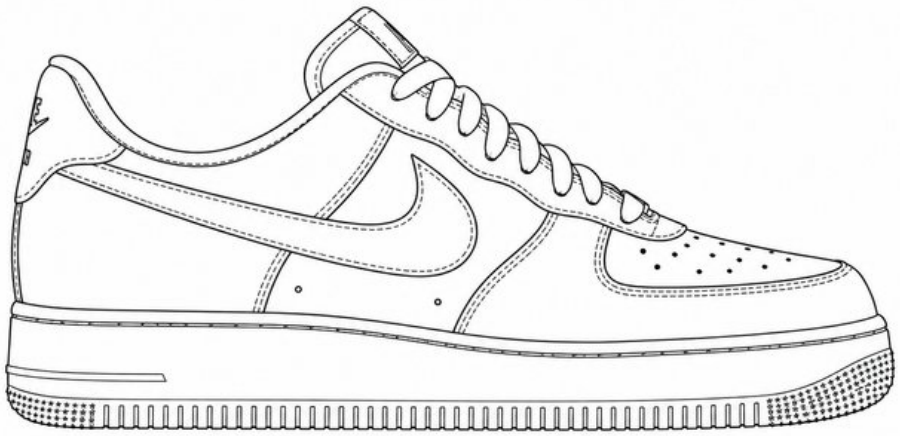
OUTSOLE VIEW



TOP VIEW



MEDIAL VIEW

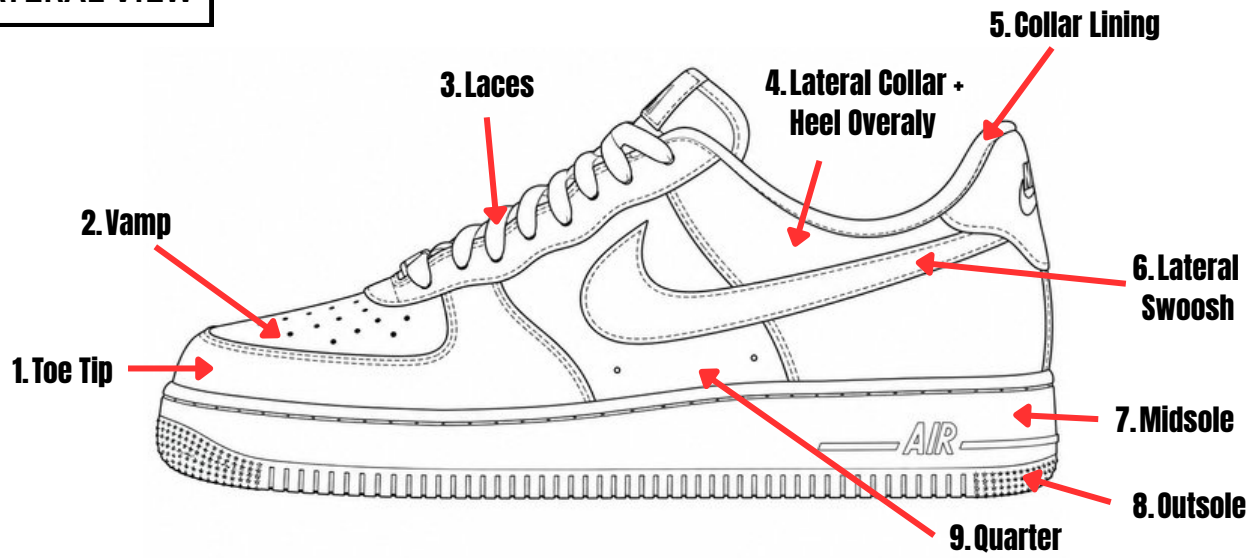


NIKE Air Force 1 Low

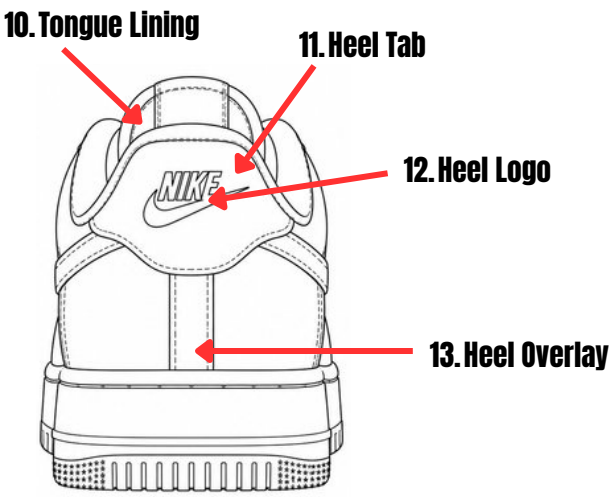
Component Callouts

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SEASON:	FALL 1983	REV:	1
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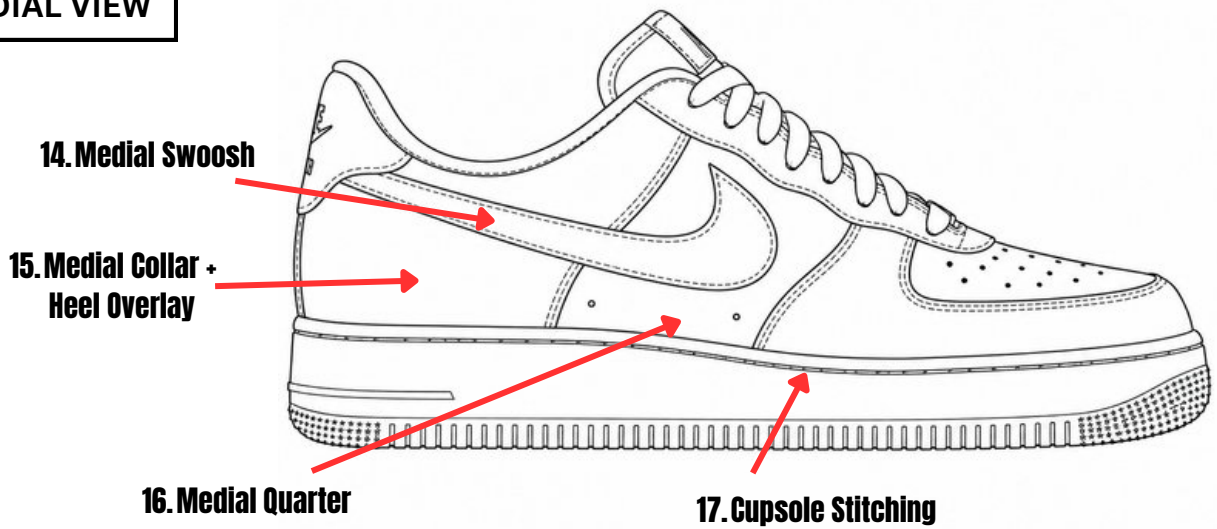
LATERAL VIEW



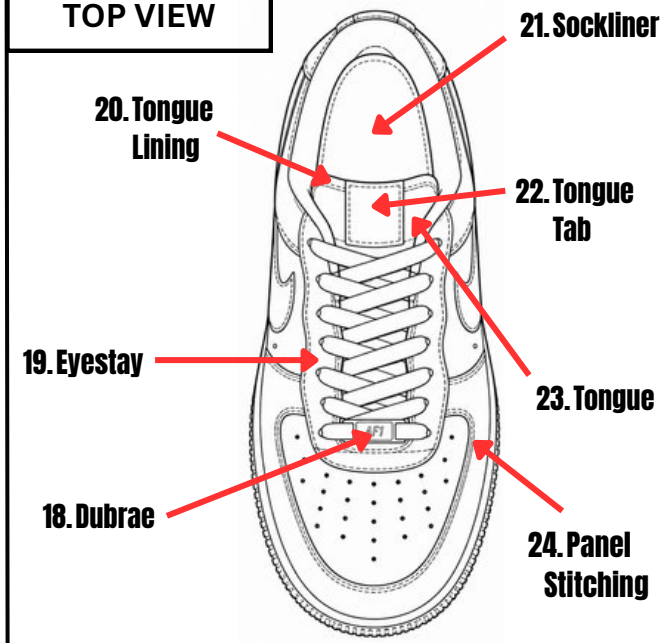
HEEL VIEW



MEDIAL VIEW



TOP VIEW

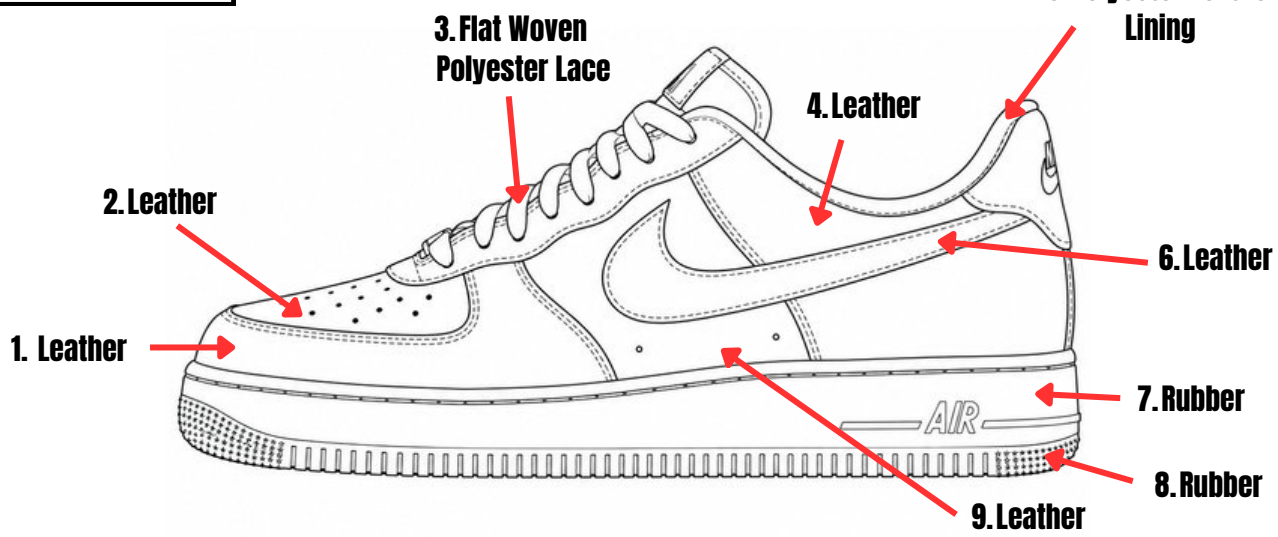


NIKE Air Force 1 Low

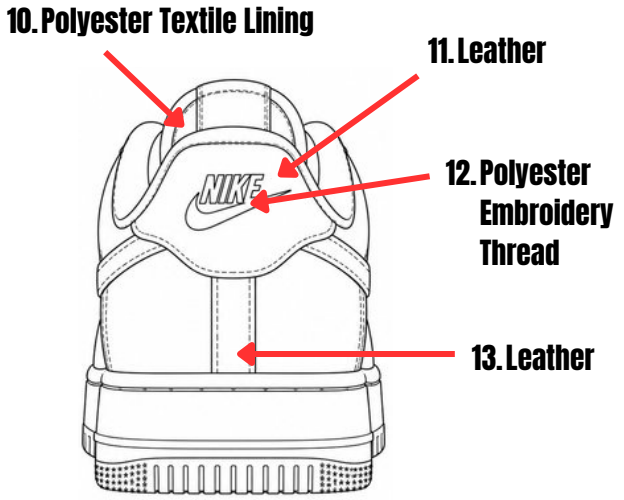
Material Callouts

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SEASON:	FALL 1983	REV:	1
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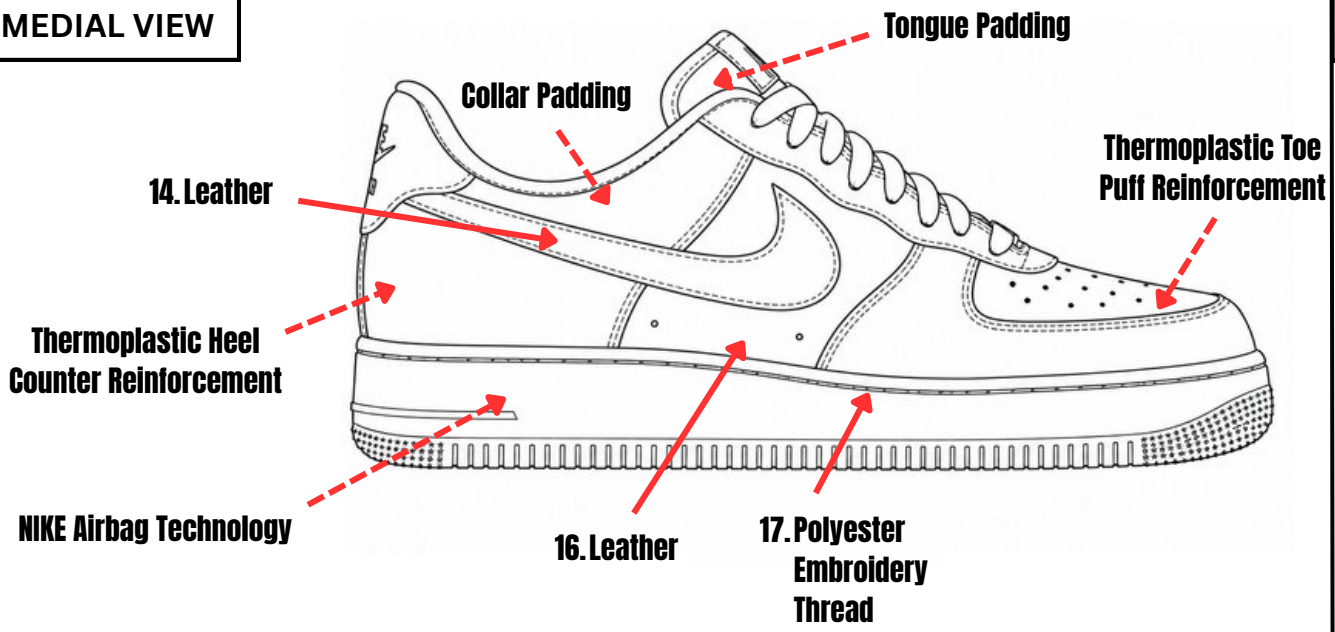
LATERAL VIEW



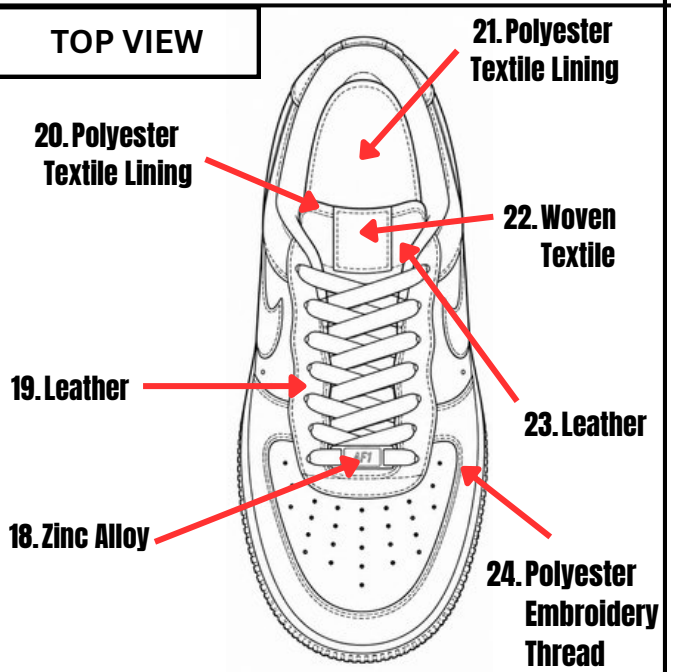
HEEL VIEW



MEDIAL VIEW



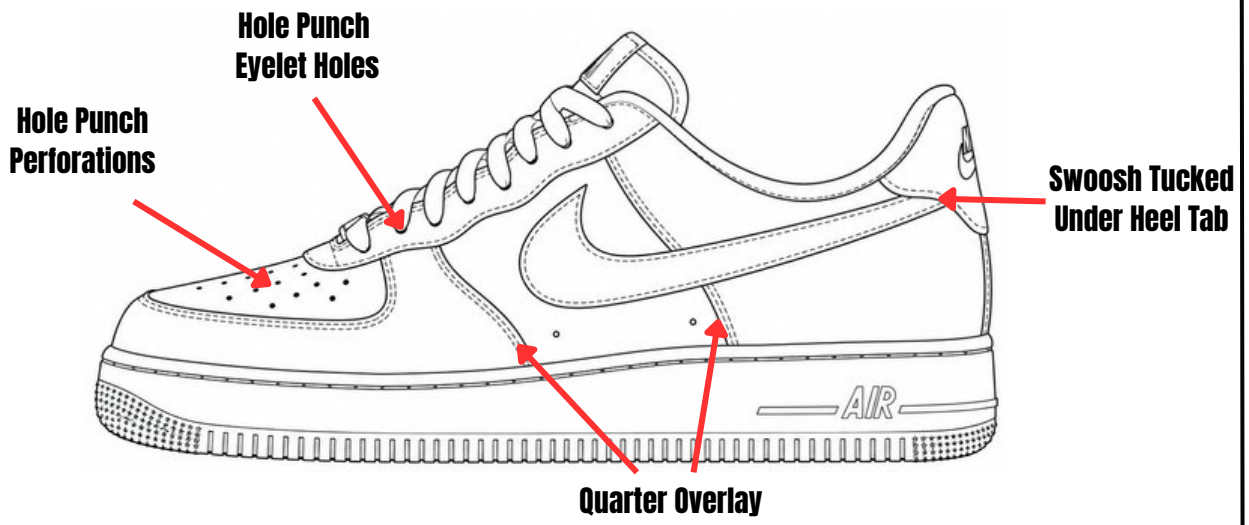
TOP VIEW

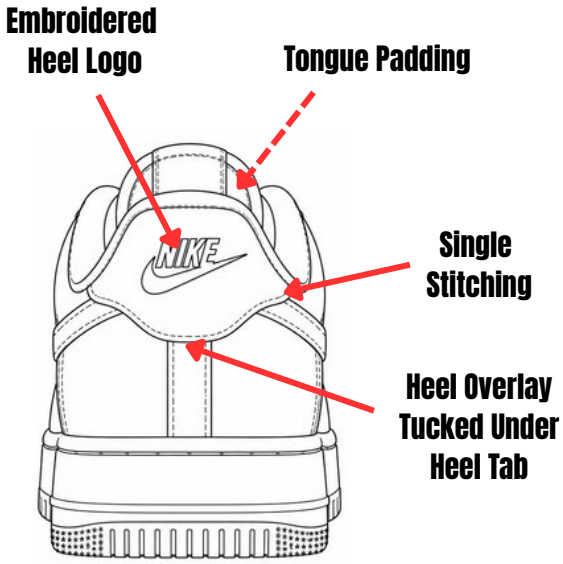


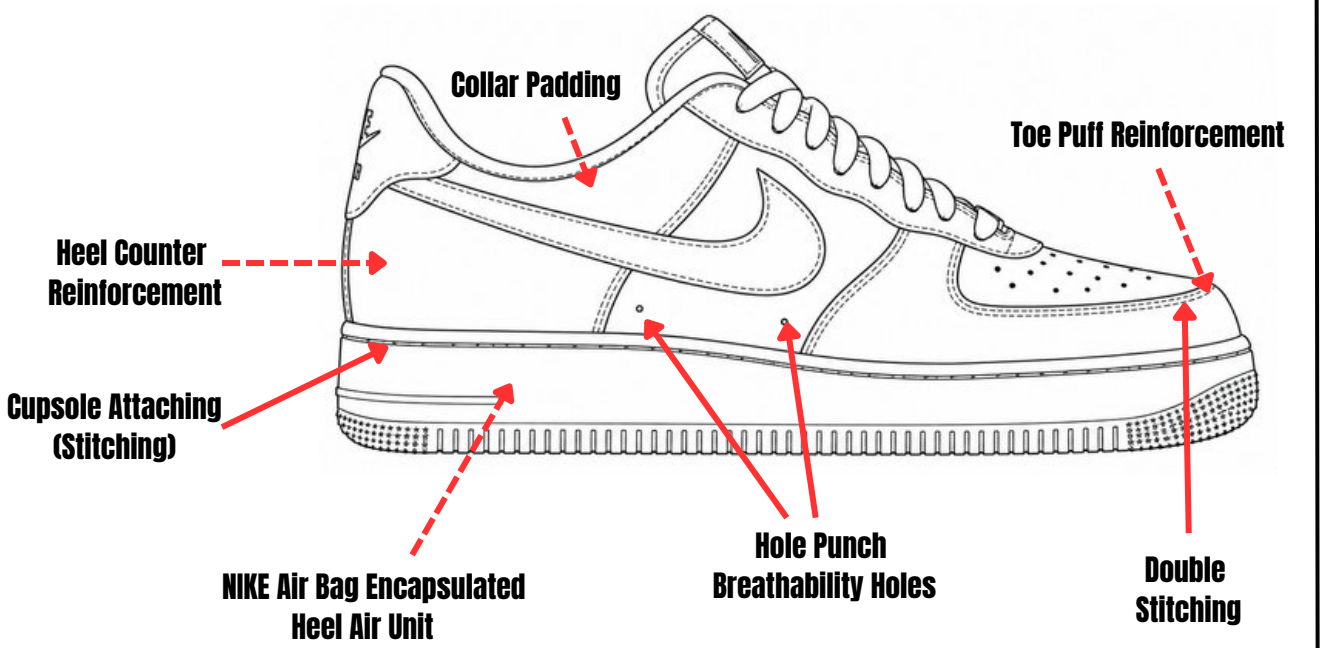
NIKE Air Force 1 Low

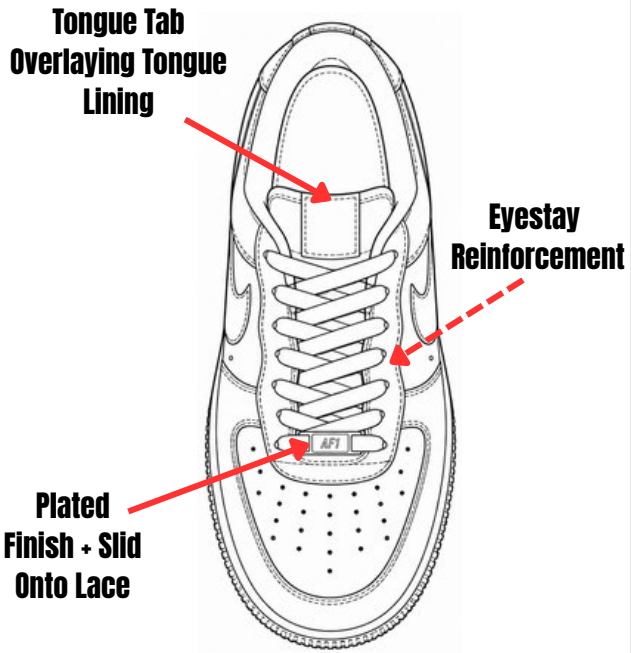
Construction Callouts

MODEL:	AIR FORCE 1 LOW	SHEET:	4 of 8
SEASON:	FALL 1983	REV:	1
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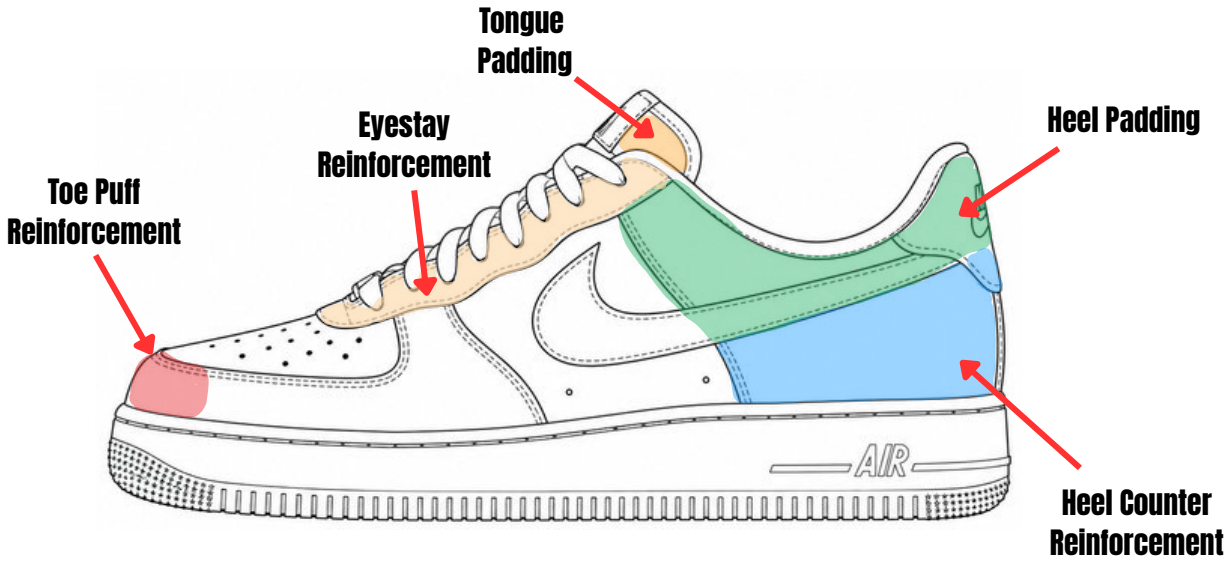


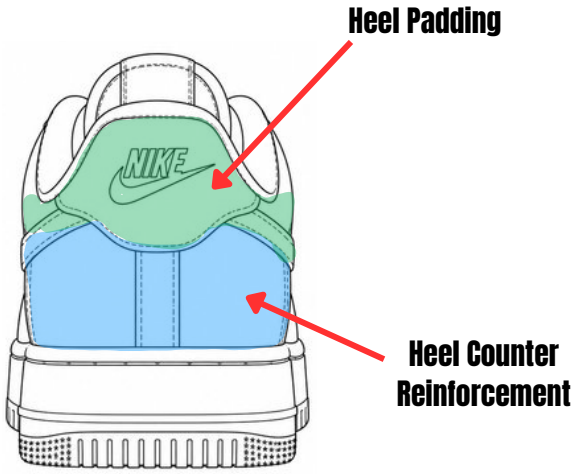


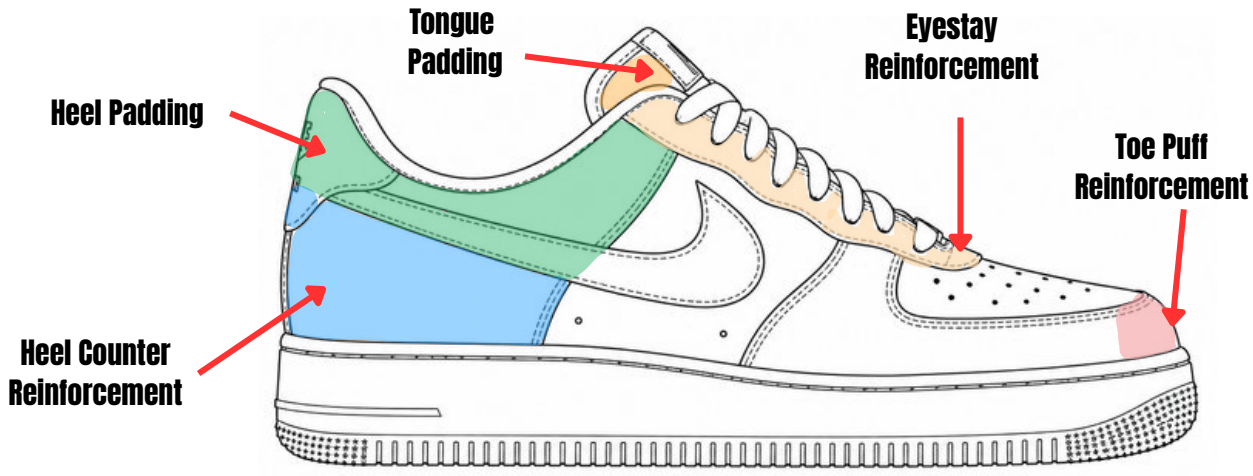
NIKE Air Force 1 Low

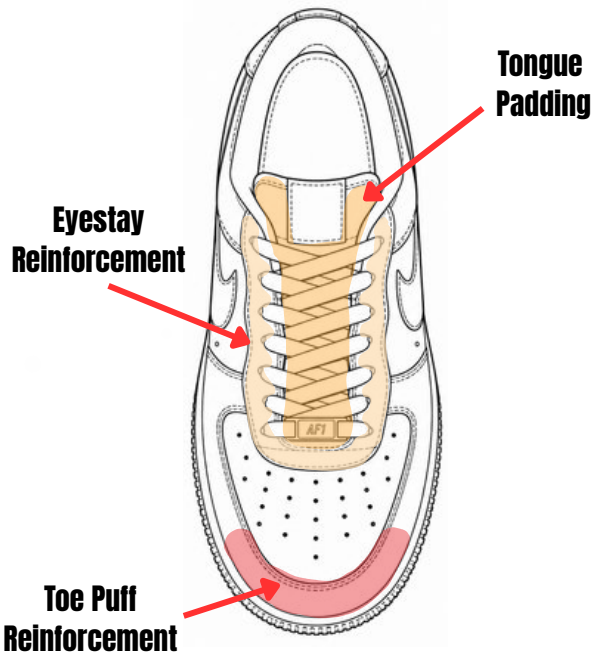
Reinforcements + Padding

MODEL:	AIR FORCE 1 LOW	SHEET:	5 of 8
SEASON:	FALL 1983	REV:	1
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Material Key

- Thermoplastic Toe Stiffner, 0.5-0.8mm
- Thermoplastic, 1.0-1.5mm
- Cotton/Polyester Woven, 8-10mm
- NonWoven Reinforcement, 0.4-0.8mm

BILL OF MATERIALS

NIKE Air Force 1 Low

Part ID	Category	Part	Material ID	CAD #	Visibility	Qty per PAIR	Purchase UOM	Cost per Unit ($)	Cost per Pair ($)	Proposed Material Spec	Color	Estimated Source / Origin
U01	Upper	Toe tip	M01	1	Visible	2	pieces	$0.21	$0.41	Cowhide full-grain, 1.2–1.5 mm	White	South Korea / Taiwan (illustrative tannery)
U02	Upper	Perforated Vamp	M01	2	Visible	2	pieces	$0.19	$0.37	Perforated cowhide full-grain, 1.2–1.5 mm	White	South Korea / Taiwan (illustrative tannery)
U03	Upper	Laces	M02	3	Visible	2	pieces	$0.07	$0.13	Cowhide full-grain, 1.2–1.5 mm	White	Taiwan / Japan (illustrative mill)
U04	Upper	Lateral Collar + Heel Overlay	M01	4, 13	Visible	2	pieces	$0.14	$0.28	Cowhide full-grain, 1.2–1.5 mm	White	South Korea / Taiwan (illustrative tannery)
U05	Upper	Collar Lining	M03	5	Visible	0.15	m²	$1.00	$0.15	Cowhide full-grain, 1.2–1.5 mm	White	Taiwan / Japan (illustrative mill)
U06	Upper	Lateral Swoosh	M01	6	Visible	2	pieces	$0.09	$0.18	Cowhide full-grain, 1.2–1.5 mm	White	South Korea / Taiwan (illustrative tannery)
U07	Upper	Lateral Quarter	M01	9	Visible	2	pieces	$0.15	$0.30	Cowhide full-grain, 1.2–1.5 mm	White	South Korea / Taiwan (illustrative tannery)
U08	Upper	Tongue Lining	M03	10, 20	Partly hidden	0.12	m²	$1.08	$0.13	Cowhide full-grain, 1.2–1.5 mm	White	Taiwan (illustrative converter)
U09	Upper	Heel Tab	M01	11	Visible	2	pieces	$0.09	$0.17	Full-grain leather, 0.9–1.2 mm	White	South Korea / Taiwan (illustrative tannery)
U10	Upper	Heel Logo Embroidery	M04	12	Visible	8	m	$0.01	$0.06	Full-grain leather, 0.9–1.2 mm	White	Taiwan / Japan (illustrative thread mill)
U11	Upper	Medial Swoosh	M01	14	Visible	2	pieces	$0.09	$0.18	Cowhide full-grain, 1.2–1.5 mm	White	South Korea / Taiwan (illustrative tannery)
U12	Upper	Medial Collar + Heel Overlay	M01	15	Visible	2	pieces	$0.14	$0.28	Polyester knit lining	White	South Korea / Taiwan (illustrative tannery)
U13	Upper	Medial Quarter	M01	16	Visible	2	pieces	$0.15	$0.30	PU foam, 4–8 mm	White	South Korea / Taiwan (illustrative tannery)
U14	Upper	Metal Dubrae	M05	18	Optional	0	pieces	$0.00	$0.00	Woven textile label	Silver	Taiwan (illustrative hardware supplier)
U15	Upper	Eyestay	M01	19	Visible	2	pieces	$0.10	$0.20	Polyester knit, 120–180 gsm	White	South Korea / Taiwan (illustrative tannery)
U17	Upper	Tongue Tab	M04	22	Visible	2	pieces	$0.04	$0.07	PU foam, 6–10 mm	White	Taiwan / Japan (illustrative mill)
U18	Upper	Tongue	M01	23	Visible	2	pieces	$0.10	$0.20	Cowhide full-grain, 1.2–1.5 mm	White	South Korea / Taiwan (illustrative tannery)
U19	Upper	Upper Stitching	M06	24	Visible	35	m	$0.01	$0.35	Polyester embroidery thread, 40 wt	White	Taiwan / Japan (illustrative thread mill)
U20	Upper	Tongue Padding	M07	—	Hidden	0.08	m²	$1.13	$0.09	Nonwoven reinforcement, 0.4-0.8 mm	Natural	Taiwan (illustrative foam supplier)
U21	Upper	Collar Foam Padding	M07	—	Hidden	0.12	m²	$0.92	$0.11	Cotton/polyester woven, 8–10 mm	Natural	Taiwan (illustrative foam supplier)
U22	Upper	Toe Puff Reinforcement	M08	—	Hidden	2	pieces	$0.08	$0.16	Thermoplastic toe stiffener, 0.5–0.8 mm	Natural	South Korea (illustrative component supplier)
U23	Upper	Eyestay Reinforcement	M07	—	Hidden	0.02	m²	$1.13	$0.02	Nonwoven reinforcement, 0.4-0.8 mm	Natural	Taiwan (illustrative foam supplier)
U24	Upper	Heel Counter Reinforcement	M09	—	Hidden	2	pieces	$0.09	$0.18	Thermoplastic heel counter, 1.0–1.5 mm	Natural	South Korea (illustrative component supplier)
U25	Upper	Eyestay Backing	M10	—	Hidden	2	pieces	$0.04	$0.08	Metal lace tag	Natural	South Korea (illustrative component supplier)
U26	Upper	Upper Assembly Adhesive	M11	—	Hidden	18	g	$0.01	$0.18	Bonded nylon thread, ticket 40	—	Japan / South Korea (illustrative chemical supplier)
I01	Interior	Sockliner	M03	21	Visible	2	pieces	$0.16	$0.32	Footwear assembly adhesive	White	Taiwan (illustrative converter)
I02	Interior	Insole / lasting board	M12	—	Hidden	2	pieces	$0.07	$0.14	PU/EVA foam, textile facing	Natural	South Korea (illustrative assembly supplier)
I03	Interior	Sockliner logo print	M13	—	Visible	2	prints	$0.02	$0.04	Nonwoven insole board	Black	South Korea (illustrative print supplier)
B01	Bottom	Midsole	M14	7	Visible	2	pieces	$0.77	$1.54	Footwear-grade print ink	White	South Korea / Taiwan (illustrative compounder)
B02	Bottom	Outsole	M14	8	Visible	2	pieces	$0.68	$1.36	Molded rubber compound	White	South Korea / Taiwan (illustrative compounder)
B03	Bottom	Sidewall Stitching	M06	17	Visible	12	m	$0.01	$0.14	Abrasion-resistant rubber	White	Taiwan / Japan (illustrative thread mill)
B04	Bottom	Encapsulated Nike Air-Sole unit	M15	—	Hidden	2	pieces	$0.35	$0.70	Encapsulated Nike Air-Sole unit	—	Nike-controlled Air-Sole supply (origin unverified)
B05	Bottom	Sole bonding adhesive / primer	M16	—	Hidden	50	g	$0.00	$0.18	Heavy bonded nylon/polyester thread	—	Japan / South Korea (illustrative chemical supplier)

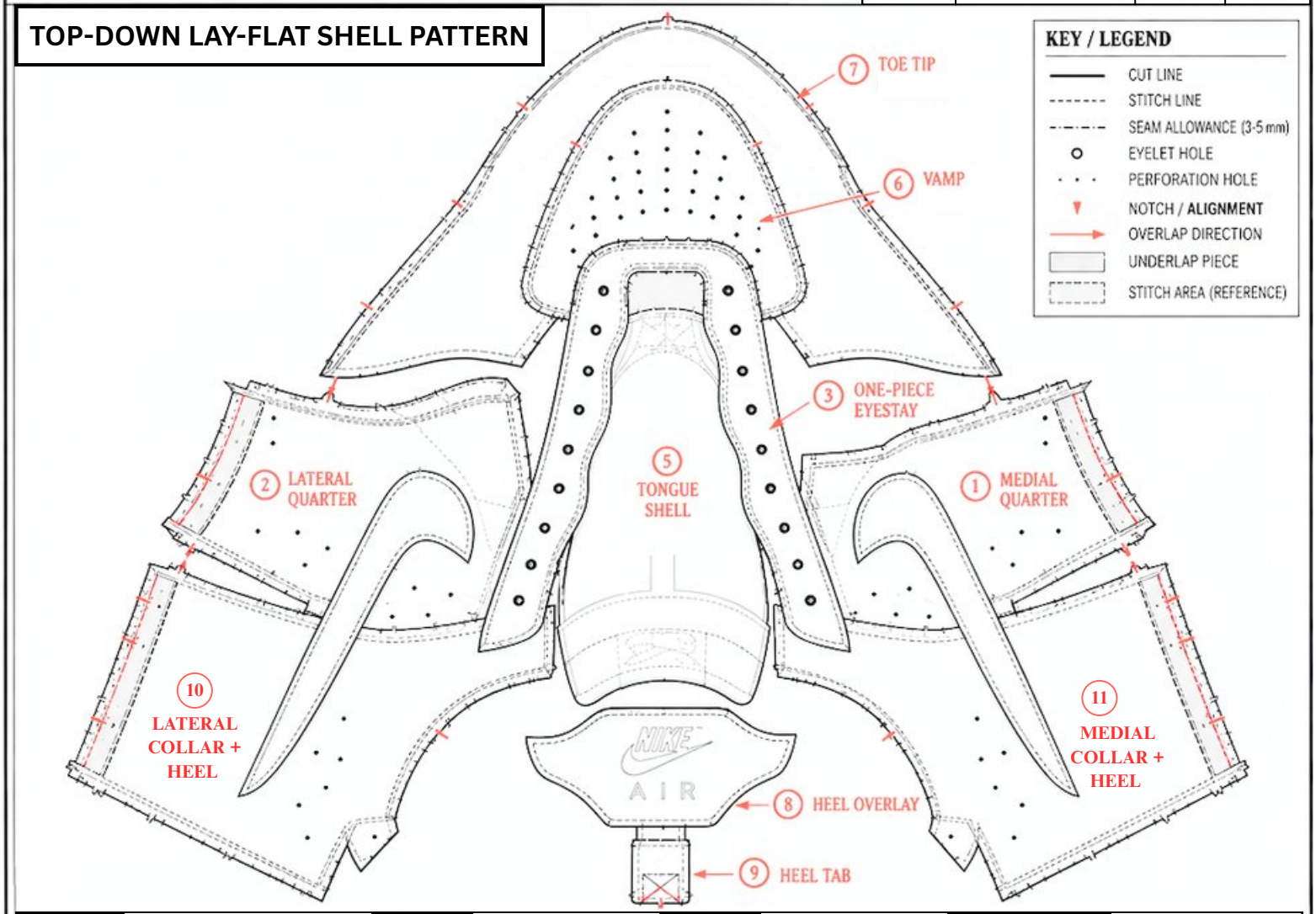
TOTAL MATERIALS	$9.00

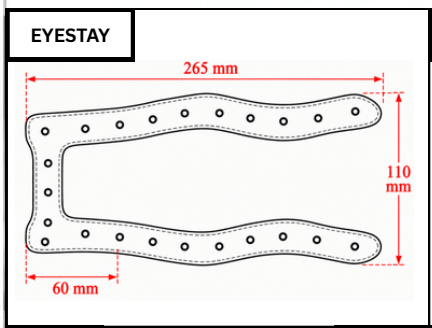
NIKE AIR FORCE 1 LOW

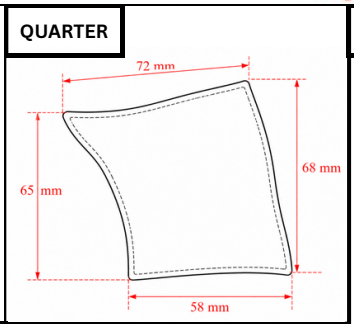
SHELL PATTERN

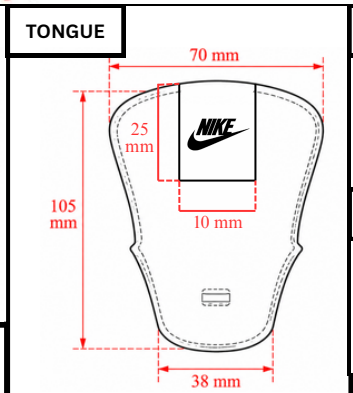
MODEL:	AIR FORCE 1 LOW	SHEET:	6 of 8
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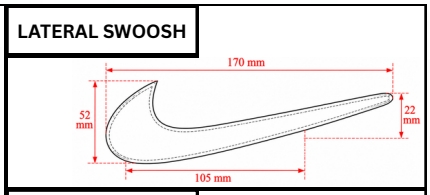
TOP-DOWN LAY-FLAT SHELL PATTERN

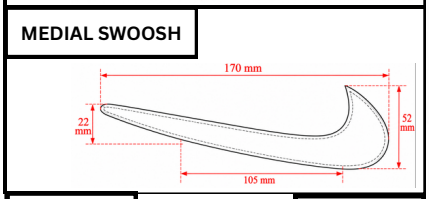


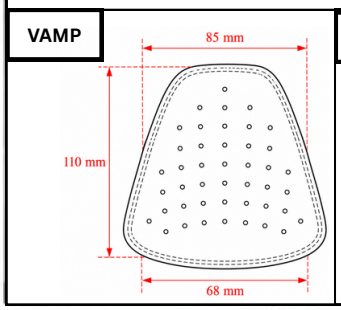


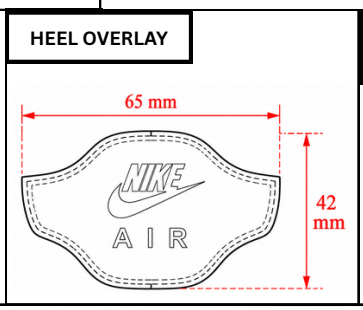


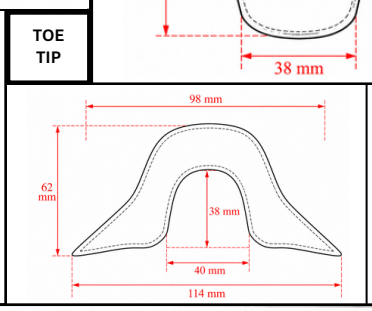


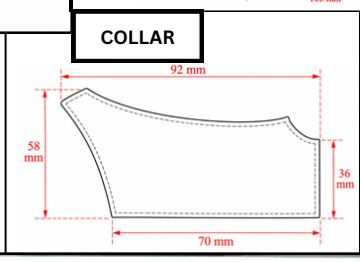


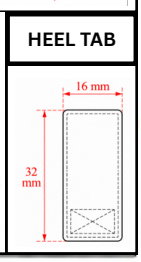












NOTES:

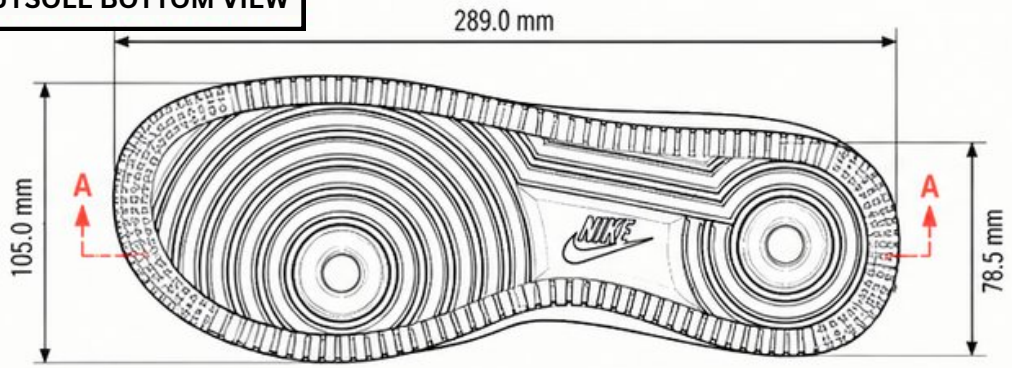
1. All dimensions shown are in mm.
2. Assembled lay-flat view shoes overlap relationships.
3. Dimensions are accurate, but sizing is not to scale.

NIKE Air Force 1 Low

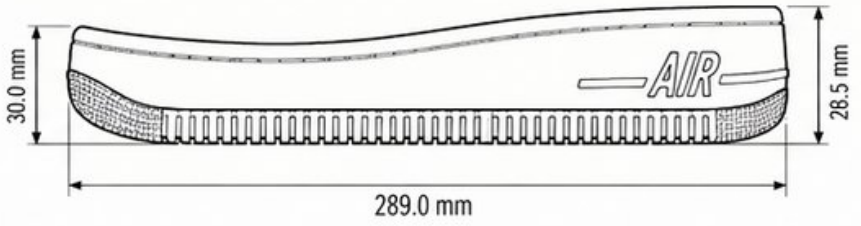
Tooling Drawing

MODEL:	AIR FORCE 1 LOW	SHEET:	7 of 8
SEASON:	FALL 1983	REV:	1
SIZE:	U.S. M9	DATE:	2/6/82

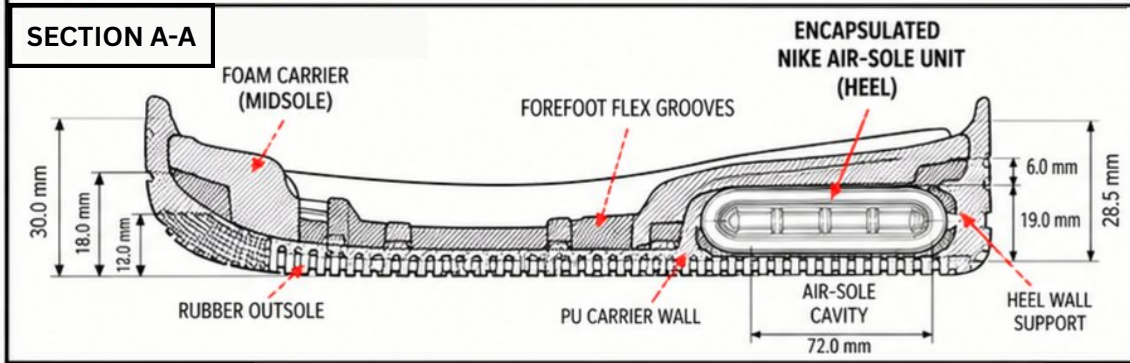
OUTSOLE BOTTOM VIEW



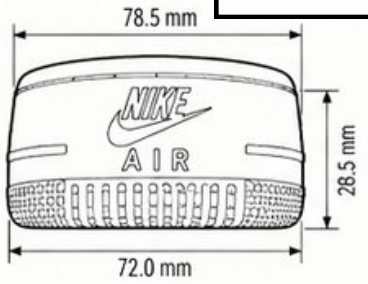
OUTSOLE SIDE VIEW



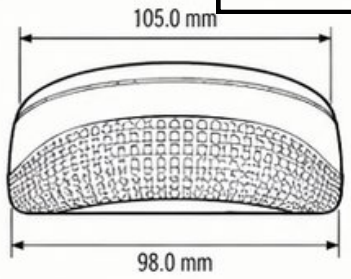
SECTION A-A



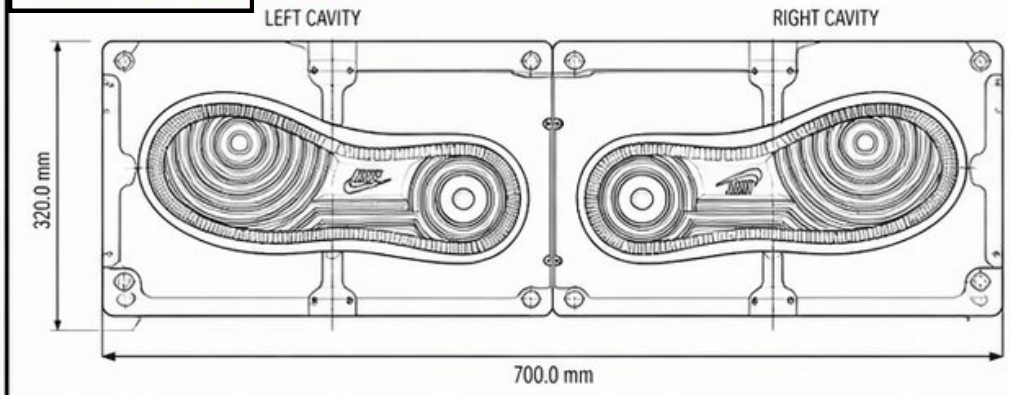
HEEL DETAIL



TOE DETAIL



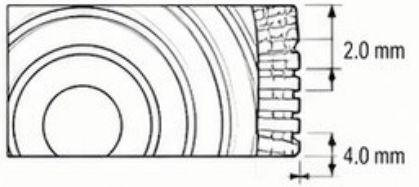
TOOLING LAYOUT



TOOLING SPECIFICATIONS

TOOL TYPE	Compression Mold
MOLD MATERIAL	P20 Steel (HRC 28–32)
CAVITY	2 Cavities (Left / Right)
PART MATERIAL	Rubber Cupsole (Solid)
PROCESS	Compression Molding
SHRINKAGE	1.5% (Allow in tooling)
FINISH	VDI 27–30 (Texture per sample)
EJECTOR TYPE	Pin Ejectors (Non-Visible Area)
ALIGNMENT	Keyed Registration
TOOL LIFE	200,000+ Pairs

TREAD PATTERN DETAIL



MIDSOLE LOGO DETAIL



NOTES:

1. All dimensions are for U.S. Men's Size 9. Allow 1.5% shrinkage in tooling.
2. Tread pattern, logo and texture match approved sample.
3. Final tooling design subject to factory review and DFM validation.
4. Tooling to be based on existing Air Force 1 cupsole specification.

NIKE AIR FORCE 1 LOW

BUDGET/COST BREAKDOWN

MODEL:	AIR FORCE 1 LOW	SHEET:	8 of 8
SEASON:	FALL 1983	REV:	1
SIZE:	U.S. M9	DATE:	2/6/82

RETAIL PRICE

$65.00

TARGET FOB / PAIR

$13.00

ESTIMATED COST PER COMPLETE PAIR

NO.	COST ELEMENT	SCOPE	$/PAIR	% FOB
01	UPPER MATERIALS	Leather upper, laces and trims	$4.60	35.4%
02	INTERIOR MATERIALS	Sockliner and lasting components	$0.48	3.7%
03	BOTTOM UNIT	Midsole, outsole and Air-Sole	$3.92	30.2%
04	PACKAGING	Box, tissue and identification	$0.50	3.8%
05	LABOR	Cutting, stitching and assembly	$1.75	13.5%
06	OVERHEAD	Factory operating allowance	$1.00	7.7%
07	FACTORY PROFIT	Manufacturer profit allowance	$0.75	5.8%
MATERIALS AND PACKAGING				**$9.50**
LOP (LABOR + OVERHEAD + PROFIT)				**$3.50**
TOTAL ESTIMATED FOB / PAIR				**$13.00**
COSTING BASIS	BOM Materials $9.00 + packaging $0.50 + LOP $3.50 = $13.00 FOB			

NIKE Air Force 1 Low

18-Month Workback Calendar

THREE KEY REVIEW GATES

GATE 1	GATE 2	GATE 3
May 21, 1982	**July 23, 1982**	**November 24, 1982**
Full product scrutiny	Wear-test authorization	Design confirmation

TARGET DATE	FACTORY ACTION / DELIVERABLE
February 6, 1982	Brief issued / factory development kickoff
March 26, 1982	Pullover review — shell pattern and appearance only
May 21, 1982	**GATE 1 — complete sample scrutiny**
July 23, 1982	**GATE 2 — fit review and wear-test approval**
August 16 – September 24, 1982	Wear test — no sample changes during testing
October 1, 1982	Wear-test readout / authorize revisions
November 24, 1982	**GATE 3 — design confirmation**
February 4, 1983	Final BOM, costing and sourcing approval
March 17, 1983	Size-set / commercialization approval
April 15, 1983	Pre-production sample approval / materials ready
April 29, 1983	Pilot production and quality approval
June 18, 1983	Bulk production complete / final inspection
July 23, 1983	Freight, customs and U.S. inventory receipt
August 6, 1983	**FALL 1983 TARGET RETAIL LAUNCH**

NIKE Air Force 1 Low

Wear Test Plan

01 TEST OVERVIEW

TEST WINDOW	August 16 - September 24, 1982
SAMPLE ALLOCATION	16 pairs requested (12 test + 4 reserve / replacement)
TEST GROUP	Competitive basketball players; recruited by Wear Test Analyst
PRODUCT	AF1 Low - Gate 2 approved sample; men's sample-size run

02 WEAR TEST SEQUENCE

DATES	TEST TYPE	PURPOSE
Aug 16-20, 1982	**Fit & Feel**	Brief fit, pressure and comfort review
Aug 23-27, 1982	**Dynamic**	Court cuts, stops, jumps; triad observes
Aug 30-Sep 10, 1982	**Short Term**	Player feedback and performance issues
Aug 16-Sep 24, 1982	**Long Term**	Durability, outsole wear and cushioning

03 EXECUTION & CONTROL

PRE-TEST RELEASE	Only approved, comfortable Gate 2 product enters external testing.
PROTOTYPE SECURITY	Use understated, unbranded / low-visibility test color where needed.
TEST ADMINISTRATION	Analyst selects testers and issues wear logs; no unapproved modifications.
OBSERVATION	Designer + Developer + Analyst attend dynamic assessment (triad).
CHANGE FREEZE	Do not change tested construction during wear-test period.

04 REVIEW / DECISION

September 27 - October 8, 1982

Analyst consolidates fit, performance, durability and failure feedback.
Review findings before authorizing revisions. Gate 3: November 24, 1982.

NIKE Air Force 1 Low
Launch Email

FROM: aidan.copeland@nike.com
TO: Factory Product Development Team
DATE: February 6th, 1982
SUBJECT: NIKE Air Force 1 Low - Fall '83 - *TECH PACK RELEASE*

Hello Team,

Please proceed with development of the NIKE Air Force 1 Low for Fall of 1983. The initial tech pack is released for factory review and I have attached it to this email. Please identify any questions you may have before preparing samples. Can't wait to work together on this project!

MODEL OVERVIEW
Season: Fall 1983
Retail/TFOB: $65/$13
Forecast: 25,000 pairs (estimate)
Sizes: Men's U.S. 7-13
Sample Size: U.S. M9

RELEASED TECH PACK
Marketing brief, CAD drawings, component + material + construction callouts, Bill of Materials, Shell Pattern, Tooling Drawing, Preliminary CBD, Work-back Calendar and Wear Test Plan are included.

PRODUCT HEROES
Low-cut leather basketball upper; encapsulated heel Nike Air-Sole cushioning; durable rubber bottom with circular pivot traction; secure heel fit and large Swoosh branding.

PRODUCT CHALLENGES
Upper fit to last; heel retention and collar comfort; cost; consistent panel stitching; Air-Sole placement and bottom unit assembly; Leather yield under wear.

SAMPLE REQUEST + REVIEW DATES
Pullover Sample Needed in Hand By: March 19th, 1982
Gate 1 Sample Needed in Hand By: May 14th, 1982
Gate 2 Sample Needed in Hand By: July 21st, 1982
Gate 3 Sample Needed in Hand By: November 22nd, 1982

Thank you,
Aidan Copeland
Footwear Product Devlelopment | NIKE
aidan.copeland@nike.com